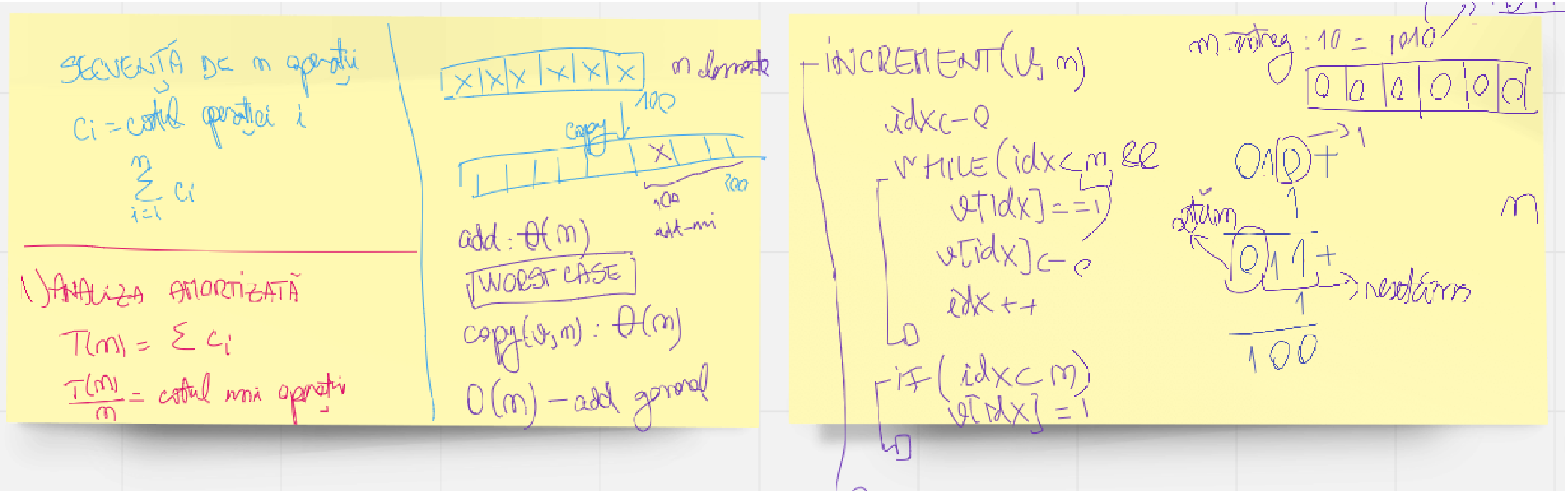


SECVENȚĂ DE n operații

c_i = costul operației i

$$\sum_{i=1}^{n} c_i$$

1) ANALIZA AMORTIZATĂ

$$T(n) = \sum c_i$$

$$\frac{T(n)}{n} = \text{costul unei operații}$$

add: $\theta(n)$

WORST CASE

copy(v, n): $\theta(n)$

$O(n)$ – add general

```
INCREMENT(v, n)
    idx <- 0
    WHILE (idx < n && v[idx] == 1)
        v[idx] <- 0
        idx++
    if (idx < n)
        v[idx] = 1
```

n întreg: 10 = 1010

n întreg : 10 = 1010

0 | 1 | 0 | 1 | 0 | 0

0 | 0 | 0 | 0 | 0 | 0

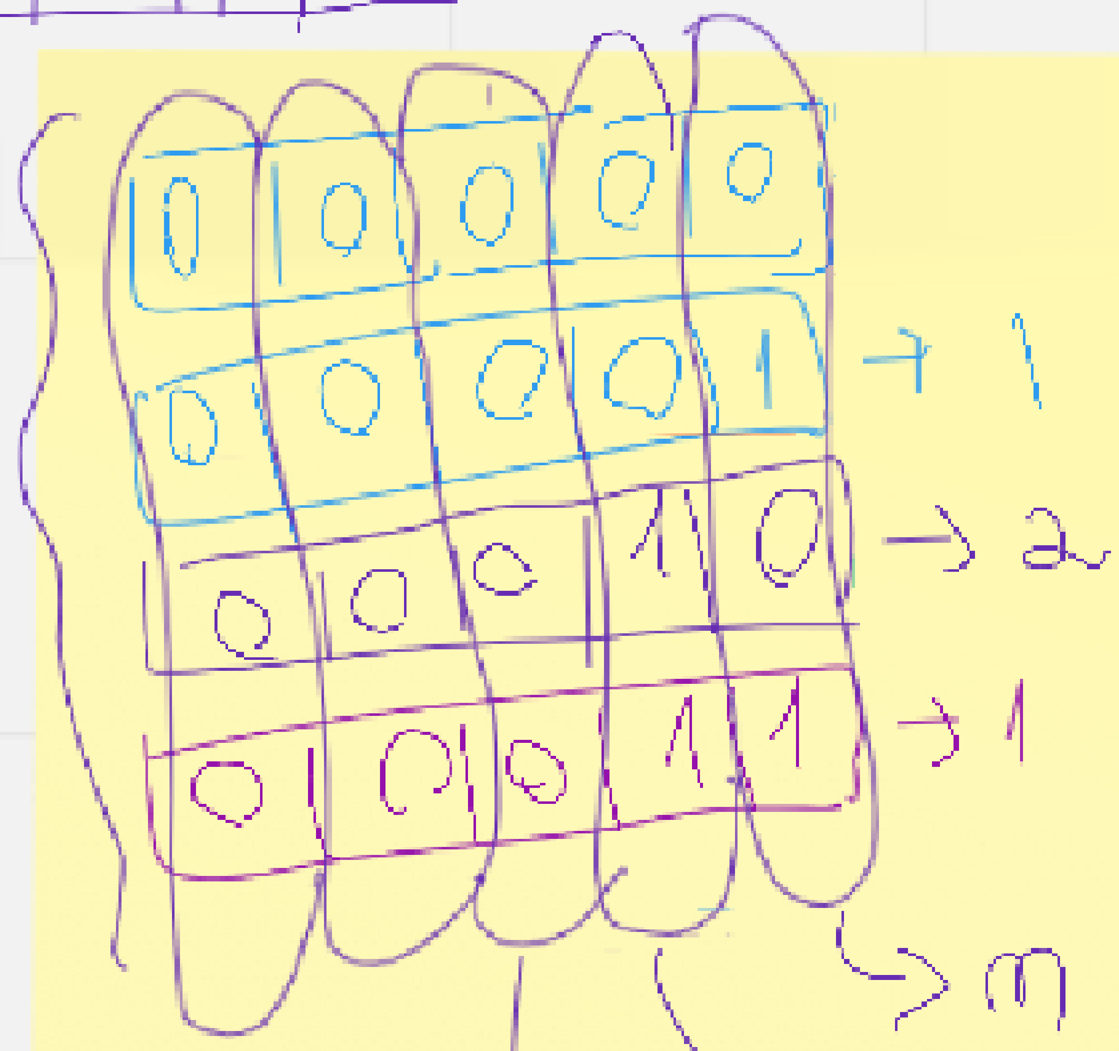
$$\begin{array}{r} 010 + \\ 1 \\ \hline 011 + \\ 1 \\ \hline 100 \end{array}$$

→ 1

ținem

resetăm

n



→ n modificări

→ $\frac{n}{2}$ -||-

→ $\frac{n}{2^2}$ -||-

$$T(n) = \sum_{i=0}^{n-1} \frac{n}{2^i} = n \sum_{i=0}^{n-1} \left(\frac{1}{2}\right)^i = n \cdot \frac{1-\overset{\to 0}{\left(\frac{1}{2}\right)^n}}{1-\frac{1}{2}} \le 2n \Rightarrow \frac{T(n)}{n} \le 2 = c_1$$

cost) = $\sum^{n}$

2) METODA CREDITELOR

$\sum \hat{c}_i \geq \sum c_i$

increment — $\boxed{1\ set}$

increment — ? reset

$c_{set} = 1$

$c_{reset} = 1$

$\hat{c}_{set} = 2$

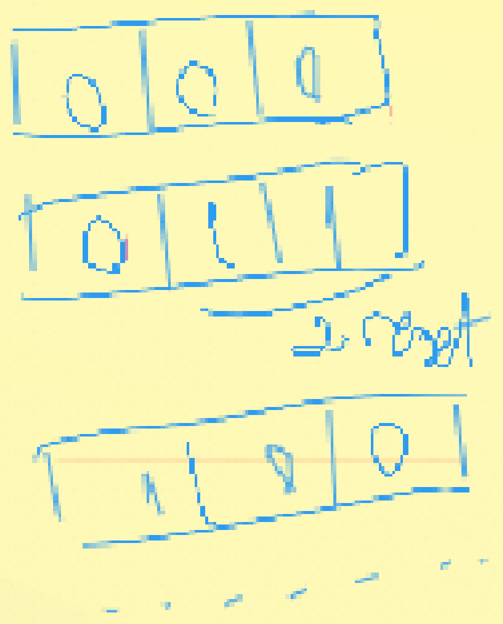
$\hat{c}_{reset} = 0$

$$\sum \hat{c}_i = \sum_{i=1}^{n} \left(\hat{c}_{set} + m_i \cdot \underset{\underset{0}{\|}}{\hat{c}_{reset}} \right) = \sum_{i=1}^{n} \hat{c}_{set} = \hat{c}_{set} \cdot n = 2n$$

$\sum c_i = 2n$

crediti $= \hat{c}_i - c_i$

$\Rightarrow 2n \geq 2n$



$\theta(n)$